Abdallah Alsammani, Ph.D.

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Interdisciplinary researcher and educator with over a decade of combined teaching and research experience in applied mathematics, mathematical biology, data science, and machine learning. Expertise in developing statistical and machine learning models for complex biological systems, public health challenges, and biomedical applications. Specialization includes mathematical modeling, high-dimensional data analysis, epidemiological modeling, and computational neuroscience. Committed to advancing translational research through data-driven, cross-disciplinary approaches in medicine, epidemiology, and health sciences.

Academic Appointments

Assistant Professor of Mathematics and Data Science (Tenure-Track) August 2025 – Present

Department of Mathematical Sciences, Delaware State University, Dover, DE

- Teach undergraduate and graduate courses in data science and applied mathematics
- Establish research program in mathematical biology, machine learning, biomedical data science, and other interdisciplinary areas
- Mentor undergraduate and graduate students in research projects
- Contribute to curriculum development and departmental strategic initiatives

Assistant Professor of Data Science August 2022 – July 2025

School of Science and Mathematics, Jacksonville University, Jacksonville, FL

- Designed and delivered courses across data science curriculum including machine learning, statistical computing, and mathematical modeling
- Established undergraduate research program supervising eight independent research projects
- Led assessment initiatives for Data Science major and minor programs
- Developed new Data Science Certificate Program
- Served on multiple search committees and university-level committees including AI Taskforce and Planning & Budget Committee

Postdoctoral Research Associate January 2022 – July 2022

Department of Infectious Diseases, University of Georgia, Athens, GA

- Conducted epidemiological modeling and statistical analysis for CDC-contracted research projects on infectious disease dynamics
- Developed computational models for disease transmission and intervention strategies
- Prepared technical reports and data-driven insights for public health stakeholders
- Supervised research group of seven Ph.D. students, coordinating collaborative projects

Postdoctoral Research Associate January 2021 – January 2022

Department of Neurosurgery, University of Nebraska Medical Center, Omaha, NE

- Developed novel computational methods for analyzing intracranial EEG data from epilepsy patients
- Created statistical frameworks for detecting and characterizing high-frequency oscillations in real-time clinical settings
- Published research on circular statistics methodology in IEEE Journal of Biomedical and Health Informatics
- Collaborated with neurosurgeons and clinical researchers on translational neuroscience projects

Graduate Teaching Assistant & Instructor of Record August 2014 – December 2020

Department of Mathematics and Statistics, Auburn University, Auburn, AL

- Served as primary instructor for undergraduate courses in calculus sequence and pre-calculus
- Supervised Auburn University Mathematics Tutoring Center for two semesters
- Received Excellence in Teaching Award (2019–2020) for outstanding instructional performance
- Mentored undergraduate students and provided comprehensive academic support

Lecturer of Mathematics and Statistics August 2012 – August 2013

Al Neelain University, Khartoum, Sudan

- Taught undergraduate courses in mathematics, statistics, and programming
- Provided academic advising and contributed to departmental curriculum development

Lecturer of Applied Mathematics August 2012 – August 2013

Academy of Engineering and Medical Sciences, Khartoum, Sudan

- Delivered courses in applied mathematics including mathematical methods and MATLAB programming

Education

Ph.D. in Applied Mathematics 2020

Auburn University, Auburn, AL

- Dissertation: *Dynamical Behavior of Nonautonomous and Stochastic HBV Infection Model*
- Focus: Mathematical modeling of infectious diseases, stochastic differential equations, dynamical systems

Postgraduate Diploma in Mathematics 2014

International Centre for Theoretical Physics (ICTP), Trieste, Italy

- Pre-doctoral program focusing on advanced mathematical methods
- Thesis: *Alexander Polynomials for Knots*

M.Sc. in Applied Mathematics 2012

African Institute for Mathematical Sciences (AIMS), Mbour, Senegal

- Thesis: *Elliptic Curve Cryptography Under Finite Field*
- Comprehensive training in mathematical modeling and computational methods

B.Sc. in Mathematics (Honors) 2009

Al Neelain University, Khartoum, Sudan

- Graduated with Outstanding Student Award
- Thesis: *Linear and Nonlinear Optimization Problems*

Research Interests

My research is inherently interdisciplinary, integrating applied mathematics, biostatistics, data science, and machine learning to study complex biological, clinical, and public health systems. I develop mathematically rigorous and computationally efficient approaches that bridge theory, data, and real-world applications in healthcare and biomedical sciences. My research directions include, but are not limited to, the following:

Mathematical and Computational Epidemiology: Deterministic and stochastic modeling of infectious diseases (e.g., hepatitis B, COVID-19, cholera), including stability analysis, identifiability, and evaluation of intervention strategies.

Machine Learning for Healthcare: Development and application of predictive models for clinical outcomes, disease risk, and health data analysis, with emphasis on interpretability and data-driven decision support.

Scientific Machine Learning: Development of physics-informed and data-driven machine learning methods for dynamical systems, including neural differential equations, hybrid modeling, and integration of mechanistic models with data for inference, prediction, and control.

Computational Neuroscience: Statistical and computational analysis of electrophysiological data, including high-frequency oscillations in epilepsy and their role in clinical diagnosis and monitoring.

Statistical Methodology: Design of robust statistical techniques for biomedical data, including circular statistics, bias correction, and analysis of complex and high-dimensional datasets.

Systems and Multiscale Modeling: Mathematical modeling of biological systems across scales, including host–pathogen dynamics, ecological interactions, and coupled within-host and population-level processes.

Selected Scholarly Publications and Research Output

Peer-Reviewed Journal Articles

1. (2026) Chen, Z., Gliske, S. V., Alsammani, A., Tyner, K., Das, S., Mutahr, M., Lin, J., Hegeman, G., Smith, G., Grayden, D. B., & Stacey, W. (2026). The influence of recording duration and vigilance state on high-frequency oscillation characterization in epilepsy. Neurology, 107(2), e218225. https://doi.org/10.1212/WNL.0000000000218225
2. (2026) Yousif, M. H., **Alsammani, A.**, Abdalla, M. H., Bashier, E. B. M., Mohammed, M. A. Y., Saeed, A. A., & Fahal, A. H. “Interpretable Machine-Learning Prognosis of Mycetoma from Routine Clinical Data.” *Transactions of The Royal Society of Tropical Medicine and Hygiene*, trag061. https://doi.org/10.1093/trstmh/trag061
3. (2026) Alflahi, A. A. E., Mohammed, M. A. Y., and **Alsammani, A.** (2026). A scalable transaction management framework for consistent document-oriented NoSQL databases. *Information Systems*, 140, 102719. doi: https://doi.org/10.1016/j.is.2026.102719.

4. (2025) **Alsammani, A.**, Ngonghala, C. N., and Martcheva, M. (2025). Impact of vaccination behavior on COVID-19 dynamics and economic outcomes. *Mathematical Biosciences and Engineering*, 22(9), 2300–2338. doi: https://doi.org/10.3934/mbe.2025084.
5. (2024) **Alsammani, A.**, Stacey, W. C., and Gliske, S. V. (2024). Estimation of circular statistics in the presence of measurement bias. *IEEE Journal of Biomedical and Health Informatics*, 28(2), 1089–1100. doi: https://doi.org/10.1109/JBHI.2023.3334684.
6. (2019) Abdulrashid, I., **Alsammani, A. A. M.**, and Han, X. (2019). Stability analysis of a chemotherapy model with delays. *Discrete and Continuous Dynamical Systems-Series B*, 24(3), 989–1005. doi: https://doi.org/10.3934/dcdsb.2019002.

Peer-Reviewed Articles (Accepted/In Press)

1. (2026) Khalafalla, M., Rueda-Benavides, J., Gransberg, D.D., & **Alsammani, A.A.** *A Methodology for Sustainable Management of Construction Equipment Fleets: Estimating Emissions and Enhancing Environmental Performance*. Accepted for publication in the Proceedings of the Construction Research Congress (CRC 2026), San Antonio, TX, 2026. (Podium Presentation).
2. (2026) Ubaka, A., **Alsammani, A.A.**, Khalafalla, M., & Regalado, D.K. *Construction Work Zone Safety: Identifying Critical Crash Factors and Patterns in Florida*. Accepted for publication in the Proceedings of the Construction Research Congress (CRC 2026), San Antonio, TX, 2026. (Podium Presentation).

Peer-Reviewed Conference Proceedings

1. (2025) Goodyear, S., & **Alsammani, A.** "Predicting Sleep Disorders Using Machine Learning: A Comparative Analysis." In *Proceedings of the International Conference on Computational Science and Its Applications (ICCSA)*, pp. 106–121.
2. (2024) Alflahi, A. A. E., Mohammed, M. A. Y., & **Alsammani, A.** "Enhancement of Database Access Performance by Improving Data Consistency in a Non-Relational Database System (NoSQL)." In *Proceedings of the International Conference on Computational Science and Its Applications (ICCSA)*, pp. 194–205.
3. (2023) **Alsammani, A.** "Mathematical Analysis of Autonomous and Nonautonomous Hepatitis B Virus Transmission Models." In *Proceedings of the International Conference on Computational Science and Its Applications (ICCSA)*, pp. 327–343.

Preprints and Manuscripts Under Review

1. (2026) **Alsammani, A.**, Johnson, M., & Elrefaei, J. "Calibrated and Interpretable Machine Learning for ICU Mortality Prediction Using First 24-Hour Clinical Data." *medRxiv*, 2026:2026–05.
2. (2026) **Alsammani, A.**, & Farah, G. "Adaptive Control and Mittag-Leffler Stability of Caputo Fractional Systems with State-Dependent Delays." *arXiv preprint arXiv:2602.07105*.
3. (2026) **Alsammani, A.A.** *A Tutorial on Symbolic Structural Identifiability Analysis of ODE Models in Julia*. arXiv preprint arXiv:2605.18910, 2026.
4. (2026) **Alsammani, A.**, Khalafalla, M., & Regalado, D. K. "Crash Severity Dynamics and Risk Determinants in Florida: An Integrated Statistical and Explainable Machine Learning Analysis." *SSRN Working Paper*, 6512551.
5. (2025) **Alsammani, A.**, Farah, G. A. M. O., Mohammed, M. A. Y., & Yavuz, M. "Cholera Transmission Dynamics with Sanitation Control Measures." *arXiv preprint arXiv:2505.08873*.
6. (2024) Mohammed, M., Mohammed, M. A. Y., **Alsammani, A.**, Bakheet, M., Hui, C., et al. "Coexistence via Trophic Cascade in Plant-Herbivore-Carnivore Systems under Intense Predation Pressure." *arXiv preprint arXiv:2408.04862*.
7. (2023) **Alsammani, A.** "Stochastic Modeling and Computational Simulations of HBV Infection Dynamics." *arXiv preprint arXiv:2308.05819*.

Conference Abstracts and Presentations

1. (2026) Das, S., **Alsammani, A.**, Stacey, W., & Gliske, S. "Circadian Rhythms in High Frequency Oscillations from Long-Term Intracranial EEG." *Artificial Intelligence in Epilepsy and Neurological Disorders Conference*.
2. (2025) Shindhelm, A., Mendez, H., **Alsammani, A.**, Hegeman, G., Stacey, W., & Gliske, S. "Patients Undergoing Intracranial EEG Monitoring Have Fragmented and Poorly Consolidated Sleep." *American Epilepsy Society (AES) Annual Meeting*.

3. (2022) Tyner, K., **Alsammani, A.**, Hegeman, G., Stacey, W., & Gliske, S. "The Impact of Intracranial EEG on Sleep Is Independent of Whether a Craniotomy Was Performed." *SLEEP Annual Meeting*.
4. (2021) **Alsammani, A.**, Gliske, S., & Stacey, W. "Effect of Sleep Stage on High-Frequency Oscillations and Artifacts." *American Epilepsy Society (AES) Annual Meeting*.

Scholarly and Educational Contributions

1. (2026) **Alsammani, A.** "Student Guidelines for Responsible Use of AI in Higher Education." *Open Science Framework (OSF)*.
2. (2020) **Alsammani, A.** *Dynamical Behavior of Nonautonomous and Stochastic HBV Infection Model*. Ph.D. Dissertation, Auburn University.

GRANTS & FUNDING

Funded Grants

Grant for Scholarship of Teaching and Learning (SoTL) 2023 – 2024
Jacksonville University
- (PI). Funded project focused on enhancing undergraduate mathematics and data science education through interactive learning modules.

Faculty Summer Research Grant 2026
Delaware State University
- (PI). Internal summer research support for interdisciplinary research in mathematical modeling, infectious disease dynamics, and data science.

Delaware INBRE Undergraduate Research Program 2026
Delaware INBRE / Delaware State University
- Faculty Mentor. Supported undergraduate biomedical and data science research projects through the DE-INBRE Summer Research Program.

Under Review / Pending

Computational STEM Pathways (CSP) 2026
Alfred P. Sloan Foundation
- (PI)(Lead Institution: Delaware State University; Partner Institution: Florida A&M University).
- Proposed project to increase undergraduate participation in computational STEM research and strengthen pathways to graduate education.

Computational STEM Pathways (CSP) 2026
Alfred P. Sloan Foundation
- (Co-PI) (Partner Institution: Hampton University) and University of Nevada, Las Vegas.
- Proposed collaborative initiative to expand computational STEM training, mentoring, and graduate-school preparation among underrepresented students.

Causal Machine Learning for Equitable STEM Achievement: Evidence from PISA 2022 and NAEP Mathematics 2026
AERA–NSF Research Grants Program
- PI. Proposed application of causal machine learning methods to identify factors that improve STEM achievement and reduce educational inequities.

Unfunded Grants

NIH R21: Secondary Analysis of Existing Datasets in Heart, Lung, Blood Diseases and Sleep Disorders 2024
National Institutes of Health (NIH)
- Co-PI. Submitted proposal focused on computational approaches to sleep disorders and biomedical data analysis.

IHER Pilot Grant Program 2026
Interdisciplinary Health Research (IHER) Center, Delaware State University
- PI. Submitted proposal entitled *Clinically Interpretable Multimodal Prediction of ICU Outcomes Using Structured and Unstructured EHR Data* focused on interpretable AI, electronic health records, and health disparities research.

Delaware INBRE Pilot Project Award 2026
Delaware INBRE Program
- PI. Submitted proposal focused on interpretable machine learning and natural language processing for predicting ICU outcomes using electronic health record data.

Teaching Experience

Delaware State University (Current) 2025 – Present
Assistant Professor, Department of Mathematical Sciences
Courses Taught:

- MTSC 821: Scientific Computations I (graduate)
- MTSC 452: Advanced Calculus II
- MTSC 571: Complex Analysis (graduate)
- MTSC 341: Probability (Online)

Jacksonville University 2022 – 2025
Assistant Professor of Data Science
Courses Developed and Taught:

- MATH 470: Machine Learning Algorithms
- MATH 270: Introduction to Data Science
- MATH 170: Data Science Foundations
- MATH 481WS: Capstone Research Project
- MATH 420: Linear Algebra II
- MATH 331: Differential Equations
- MATH 315: Probability
- MATH 240: Calculus III
- MATH 205: Elementary Statistics

Additional Responsibilities:

- Supervised eight independent undergraduate research projects (MATH 487RI)
- Mentored student internships (MATH 490)
- Developed innovative pedagogical approaches incorporating active learning and computational tools

Auburn University 2014 – 2020
Graduate Teaching Assistant & Instructor of Record
Courses Taught as Primary Instructor:

- Calculus III, Calculus II, Calculus I
- Pre-Calculus, College Algebra

Courses Supported as Teaching Assistant:

- Introduction to Statistics
- Linear Algebra
- Differential Equations

Additional Responsibilities:

- Supervised Auburn University Mathematics Tutoring Center (two semesters)
- Provided comprehensive academic support to hundreds of students

- Recognized with Excellence in Teaching Award (2019–2020)

Al Neelain University & Academy of Engineering and Medical Sciences 2010 – 2013
Lecturer, Sudan

- MATLAB Programming
- Introduction to Statistics
- Real Analysis
- Topology
- Number Theory
- Mathematical Methods
- Abstract Algebra

Research Mentoring & Supervision

Current Students Delaware State University

- **Fiona A. Ochieng** (M.S. Student), Delaware State University, Spring 2026–Present.
- **Merasia M. Johnson** (B.S. Student), Delaware State University, Spring 2026.

Undergraduate Research Supervision Jacksonville University

- **Sarah Goodyear —** Spring 2025. Independent research on statistical learning methods.
- **Holly Gallup —** Spring 2025. Data analysis and visualization for animal shelter outcomes.
- **Ann Ubaka —** (Fall 2024 & Spring 2025. Predecting Car crashes using machine learning techniques.
- **Sarah Goodyear —** Fall 2024. Internship supervision (data analytics in industry).
- **Sarah Wehrung —** Fall 2023. Road accident data analytics and visualization.
- **Rhea White —** Fall 2023. Introduction to data science and prediction algorithms in R.
- **Cody Thomas —** Spring 2023. NFL game prediction modeling using singular value decomposition.

In addition, I served as Academic Advisor for eight undergraduate students in the Mathematics and Data Science programs at Jacksonville University. Provided individualized guidance on degree planning, course selection, academic progress, career exploration, internship opportunities, and graduate school preparation, while mentoring students in their academic, professional, and research development.

Honors & Awards

Excellence in Teaching Award 2019 – 2020
Department of Mathematics and Statistics, Auburn University

Outstanding Undergraduate Student Award 2004 – 2009
Al Neelain University, Khartoum, Sudan

Fellowships & Scholarships

Graduate Teaching Assistantship 2014 – 2020
Auburn University (six years of full funding)

Pre-Ph.D. Fellowship in Mathematics 2013 – 2014
International Centre for Theoretical Physics (ICTP), Trieste, Italy

Master's Scholarship in Applied Mathematics 2011 – 2012
African Institute for Mathematical Sciences (AIMS), Senegal

Professional Development & Continuing Education

Biostatistics Professional Certificate Completed
Johns Hopkins University (Coursera)

- Summary Statistics in Public Health
- Hypothesis Testing in Public Health
- Simple Regression Analysis in Public Health
- Multiple Regression Analysis in Public Health

FlexStack: Python Fundamentals Certificate Completed
Georgia Institute of Technology

- Python Fundamentals 1: Foundations and Syntax (INTD 4001P)
- Python Fundamentals 2: Data Structures and Modules (INTD 4002P)
- Python Fundamentals 3: Web Technologies and Data Processing (INTD 4003P)

Google Data Analytics Professional Certificate In Progress
Google/Coursera (5 of 8 courses completed)

Academic Service

University-Level Service Jacksonville University

- Member, Planning and Budget Committee (2024–2025)
- Member, Artificial Intelligence Taskforce Committee (2023–2024)
- Multiple faculty search committees:
  - Assistant Professor in Computing Science, Davis College of Business (2023–2024)
  - Assistant Professor in Mathematical Physics (2023–2024)
  - Assistant Professor in Oceanography, Marine Science Department (2023–2024)
  - Assistant Dean, Davis College of Business (2022–2023)
  - Visiting Assistant Professor in Mathematics (2022–2023)

Departmental & College Service Jacksonville University

- Assessment Coordinator for Data Science Programs (2022–Present)
- Applied Science Committee Member, College of Arts and Sciences (2022–2023)
- Developed Data Science Certificate Program (2023)

Professional Service & Outreach Auburn University

- Event Coordinator, Regional Science Olympiad (March 2017)
- Organizing Committee Member, 52nd Spring Topology and Dynamical Systems Conference (March 2018)

Technical Skills & Competencies

Programming Languages & Software

- **Statistical Computing:** R, SAS, Python (NumPy, SciPy, Pandas, Statsmodels)
- **Mathematical Modeling:** MATLAB, Mathematica, Scilab
- **Machine Learning:** Python (scikit-learn, TensorFlow, Keras), R (caret, randomForest)
- **Data Visualization:** Python (Matplotlib, Seaborn), R (ggplot2), Tableau
- **Other Languages:** C++, SQL
- **Specialized Software:** BEAST (Bayesian evolutionary analysis)

Methodological Expertise

- **Statistical Methods:** Hypothesis testing, regression analysis, survival analysis, time series analysis, Bayesian inference, circular statistics
- **Machine Learning:** Supervised learning (classification, regression), unsupervised learning (clustering, dimensionality reduction), deep learning, model validation
- **Mathematical Modeling:** Ordinary and partial differential equations, stochastic differential equations, dynamical systems, optimal control theory
- **Data Science:** Data wrangling, exploratory data analysis, feature engineering, predictive modeling, big data analytics

Operating Systems

- Windows, Linux (Ubuntu, Mint), macOS

Professional Memberships

- American Epilepsy Society (AES)
- American Mathematical Society (AMS)

References

Available upon request.